A Federated Learning Approach to Intrusion Detection in Software-Defined Networks

Santhosh Voruganti
Dept. of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, India
vsantosh_it@cbit.ac.in

Sairam Utukuru
Dept. of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, India
usairam_it@cbit.ac.in

Sravyasri Shambuni
Dept. of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, India
sravyasrinivas788@gmail.com

Yeshwanth Valishetti
Dept. of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, India
yeshwanthvalishetti20@gmail.com

Hruthwik Sai Mote
Dept. of Information Technology
Chaitanya Bharathi Institute of Technology
Hyderabad, India
hruthwik05@gmail.com

***Abstract*—Network Intrusion Detection Systems are essential in the fields of Software and Hardware. In Networking, the use of Software-Defined Networks is rapidly increasing within Information Technology to manage traffic in data centers while enhancing security. However, due to their centralized management, there is a significant risk of data breaches. Securing SDNs is crucial for both data centers and media. Numerous existing models implement IDS using machine learning algorithms with various datasets, but a major drawback is that the training data provided by users is often insecure, as it can be accessed by others. This is where traditional algorithms fall short. Federated Learning offers an improved approach to implementing these detection models by enhancing the security of user-provided data. It employs collaborative learning, where individual models from clients are aggregated, and the average is taken. This concept of integrating Federated Learning with the base algorithm safeguards user privacy. The proposed model stands out in terms of the number of attacks detected and in alerting users when an intrusion is identified.**

***Index Terms*—Federated Learning,Aggregation,Software Defined Networks,Collaborative Learning**

I. Introduction

With the enormous amount of network traffic, securing Software-Defined Networks (SDNs) has become increasingly difficult, with centralized SDN controllers being especially susceptible to attacks that can bring down the entire network [1]. Conventional Intrusion Detection Systems (IDS) based on deep learning models tend to be plagued by privacy, scalability, and security issues because of centralized data gathering [2]. To tackle such challenges, we suggest employing Federated Learning (FL) to facilitate intrusion detection in SDNs, enabling individual distributed SDN nodes to learn a customized model from their local traffic without violating data privacy. FL has also proven to be beneficial in several areas to strengthen security and scalability through the decentralization of the training process, as indicated in [3] and [4]. Earlier research in machine learning, i.e., Decision Trees and Logistic Regression, has been utilized for SDN IDS, but the models tend to fail with big data and complex types of attacks [5]. Deep learning methods like CNNs and LSTM networks have also been considered for advanced attack detection [6], but they are still susceptible to privacy threats in centralized environments. Our model advances these efforts further by combining Federated Learning with deep learning methods, enhancing SDN security and offering a privacy-preserving and more scalable approach to intrusion detection [7].

Conventional Machine Learning (ML) algorithms tend not to perform well with huge volumes of network traffic data. They can become incapable of optimal convergence because of the complexity of high-dimensional data. To work around this shortcoming, the model in question uses Deep Learning (DL) algorithms, including CNN,ANN.

Rather than having a centralized training method, in which all the client information is gathered in one place (and raising privacy issues), the model presented here utilizes Federated Learning (FL): Every client trains a model locally, keeping data private. The trained models are then combined using methods such as: Federated Averaging (FedAvg) – A basic algorithm that averages the client models to create a global model.

Contrary to current Federated Learning-based Intrusion Detection Systems (IDSs), in which all the clients learn with the same model, our method introduces client-specific model training. This implies: very client learns with a different deep learning algorithm (CNN, ANN, LSTM, or Autoencoder) depending on its local dataset properties.

Current IDS models tend to categorize only a few attack types, which limits their real-world utility. The new model

classifies attacks into 14 types which are originated from 4 main classes

Every existing system of intrusion detection works on two concepts which initially started from predicting the occurrence of attack and then extended for classifying the type or nature of the attack,but there are many factors that are needed to consider before judging a model with it's accuracy.The need of considering the performance at any given situation is very important.Initially machine learning models came into implementation for identifying the attacks. The Rest of the paper is organized as follows: related work is presented in section II, Methodology in section III, Experimental resutls are in section IV, and the final conclusion is in section V.

II. Related Work

The detection of attack nature by using various machine learning algorithms was implemented in SDN.Various machine learning algorithms have been applied including Decision trees,Random forest,Logistic Regression.A simple and basic methodology is followed which is also equally important base model to be studied.This existing system which was proposed by author of [8] standardizes the data and train the data on the model then classify the type of attack.This model as trained on different machine learning algorithms and found the best average as accuracy with 0.9 which classifies 5 types of attacks including (benign).This model is suitable for less amount of data or network traffic as we have huge amount of data packets or traffic a simple machine learning model will not be suitable to handle all the huge data.This is the major finding that has considered and opted to perform our proposed model using Deep Learning Algorithms,along with this the proposed model couldn't classify a wide range of attacks as it may be due to data-set limitations so choosing a perfect dataset is also equally important.Along with this model as an extended version we have ensemble learning mechanism which was used in [9] this result in good approach for data division but this does not use any specific methodology to ensemble which can be done using Federated Learning

Similarly another existing model which follows machine learning algorithms for intrusion detection in SDN,but the main insights that can be considered from this model could be the choice of dataset regarding the particular attack in this model author Lee et al. [10] choose DDO's attacks as their main concern which used to be considered as the major threat.The existing model analyzes how the network packets will be traced and what information can be extracted from those packets such as Port numbers,IP addresses etc. The dataset that authors had chosen for the proposed model is also concerned with DDO's attack because these attacks majorly takes place in control plane where huge number of packets can be forged and sent to the target system.The major inputs that can be considered from this proposed system choice of dataset because over the years other than Distributed Denial of service attacks many more attacks started causing network intrusion so all the planes of SDN need to be considered as important as control plane.Few other models implemented in [11] which are decision trees to detect DDOs attack but again these are limited to the increase of attacks on SDN's otehr than DDO's

The concept of using Deep learning architecture is an idea where more number of network can be handled due to the layered architecture of deep learning models.The existing work [12] defines the same by using convolutional neural networks for identifying the attacks in SDN. Initially, the data cleaning process starts and another important standardizing method is using data normalization to make sure that Neural networks can clearly analyze the categorical data which are the only data considered in training. Another deep learning approach studied in [13] used LSTM to detect attacks SSH and DDO's ,in this model the concpet of choosing LSTM is an insight for the proposed model

One of the major inputs that can be considered from the existing models till now is the idea of choosing Deep Learning to work on proposed model.The next concern is security.In normal models which were studied until now train the dataset as whole and then use some part of the data for testing.Here if a person contributes his data for training then there is no security provided for that data as the same data may be exposed to another person who trains the model.For this concern the concept of Federated Learning is included as a part of implementation.The existing model [14] includes the concept of Federated Learning with Convolutional neural networks. The data is divided among the clients and individual training of the data which at last gets aggregated by Fedavg algorithm.The same concept of using Federated learning was also studied in [15] but this model used LSTM model for all clients which is not very reliable in real time data

Federated Learning is extensively using in every possible field so as in network security [16].Further implementation of the existing model achieved in increasing the number of attack types which is proposed by MubasharRaza in which the number of attacks were increased to 11-14 different classes.The implementation of Federated Learning remains same but the algorithm used for training may vary according to the needs [17].This existing model gives an overall accuracy of 0.96.But all the clients lets say number of clients are 3 then all these three clients are trained with same algorithm,so if we aggregate them then the result may be not efficient for every situation of network traffic. So,training every client with the same algorithm may not be an efficient method.So choosing different algorithms to train each client gives an efficient solution every time [18].

Summary of the related work is presented in Table 1. Highlightint the gaps in existing works and other details such as datasets.

III. Proposed Model

The NSL-KDD dataset is utilized to train and test the suggested Intrusion Detection System (IDS). For maintaining data consistency and successful model training, data preprocessing operations are performed. First, the target is dropped

TABLE I
SUMMARY OF RELATED WORK

Title	Author(s)	Year	Methodology	Dataset	Findings	Drawbacks
Machine Learning Algorithms for Enhancing Intrusion Detection Within SDN/NFV	Amina Sahibi et al.	2024	Machine learning models for SDN/NFV intrusion detection	SDN Intrusion	Enhanced detection accuracy for SDN/NFV environments	Obsolete datasets; lacks advanced attack simulations
Detection Methods for Software Defined Networking Intrusions (SDN)	Multiple authors	2023	Comparative analysis of SDN intrusion detection methods	NSL-KDD, CICIDS2017	Highlights strengths and weaknesses of existing methods	High false-positive rates;
An Efficient Federated Learning System for Network Intrusion Detection	Jianbin Li et al.	2024	Federated learning for collaborative IDS	Custom simulated datasets	Demonstrated feasibility of FL in intrusion detection with high detection performance	Communication overhead; poor handling of non-IID data distributions
Federated Learning for Privacy-Preserving Intrusion Detection in SDNs	Mubashar Raza et al.	2024	Federated learning with multiclass classification IDS	Edge-IIoTset	Improved security in SDNs with privacy-preserving FL approach	Limited to multiclass classification; potential delays in model convergence

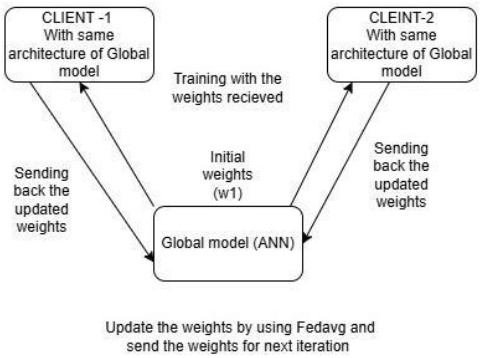

Fig. 1. Federated Learning

from the dataset and then few features like protocol-type are converted to numerical values from categorical .The dataset is further divided between multiple clients in a federated learning environment. When three clients are considered, the data is divided evenly across them, and each client gets an equal subset with a mix of both normal and attack traffic patterns. The flow is shown in Figure 1. The diagram depicts a federated learning process using an Artificial Neural Network (ANN) as the global model. In this process, there are multiple clients, illustrated here as ”CLIENT-1” and ”CLEINT-2,” each possessing the same architecture as the global model. The global model, labeled ”Global model (ANN),” initializes the process by sending its initial weights (w1) to each of the clients. Subsequently, the clients use these received weights to train their local models using their respective datasets. After training, each client sends back the updated weights to the global model. The global model then aggregates these updated weights using a technique like FedAvg (Federated Averaging) to update its own weights. This updated global model then sends its weights back to the clients for another iteration of training, effectively creating a cyclical process of distributed learning and model refinement. The note at the bottom indicates that the global model updates its weights using FedAvg and then sends the weights out for the next iteration.

A. Local Model Training

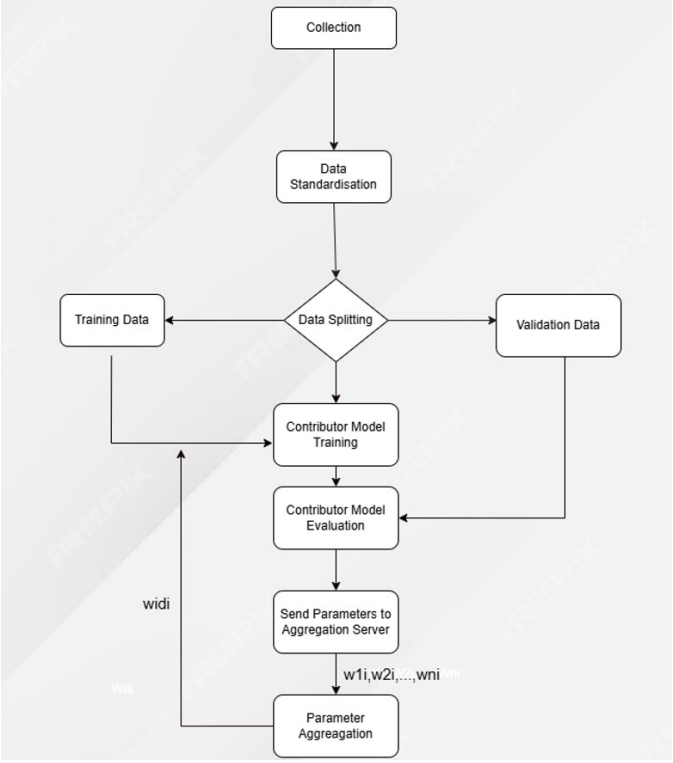

Fig. 2. Architecture Diagram

In contrast to common federated learning methods in which all the clients train based on the identical model, this work utilizes LSTM Autoencoders locally at each client. All clients train the local LSTM Autoencoder on its local dataset individually and learn effective representation for identifying anomalies. Each client’s main hyperparameters like the learning rate, number of epochs, and the dropout values are tuned individually in order to provide better performance in

the models.Because actual network conditions can differ by location, training clients with varied architectures guarantees reliability in identifying multiple types of attacks.

B. Aggregation of Local model

After local training, the new model parameters from all the clients are transferred to the global server. The global server combines all these parameters using the Federated Averaging (FedAvg) algorithm. The architectural diagram is presented in Figure 2. The diagram illustrates a machine learning workflow, most likely within a federated learning or distributed setting. It begins with a Collection stage, where raw data is gathered from various sources or contributors. Subsequently, the collected data undergoes Data Standardization, transforming it into a uniform format to ensure consistency across different datasets. Following standardization, the data is divided into two subsets: Training Data and Validation Data. The training data is utilized to train individual contributor models, while the validation data is used to evaluate their performance.

Next, each contributor trains a machine learning model using their local Training Data in the Contributor Model Training step. After training, the models are evaluated using the Validation Data in the Contributor Model Evaluation phase. Instead of sharing the raw data, each contributor sends the learned parameters of their trained model to a central Aggregation Server during the ”Send Parameters to Aggregation Server” step. The Aggregation Server receives these parameters and aggregates them to create a global model in the Parameter Aggregation stage, resulting in aggregated parameters represented as w1i, w2i, ..., wni. Finally, a feedback loop, denoted as ”widi,” connects the ”Parameter Aggregation” step back to the ”Contributor Model Training” step. This iterative process allows the global model’s aggregated parameters to update and refine the local contributor models, facilitating convergence toward a better overall solution. The FedAvg algorithm calculates a weighted average of the client models to incrementally update the global model.The global model is trained for five rounds of training, in which after every round, the new parameters are distributed to clients for additional fine-tuning. If the global model fails to converge to a best solution, further hyperparameter tuning is done, and training is re-executed to improve classification accuracy.

IV. Experimental Results

The experimental evaluation started with a baseline on the application of CNN and ANN models with accuracies of 77 and 78, respectively. These models had no consideration for data privacy, necessitating the introduction of Federated Learning. As a first step, a homogeneous Federated Learning setup was attempted by which two clients learned on equally divided data using an ANN-based model with an accuracy of 78. To improve the performance, the new method adopted is to use different models for different clients for more reliability which are LSTM Autoencoders at the client side and an LSTM model global model so that a better feature extraction process is facilitated. The summary of the performance comparison is shown in Table 2.

TABLE II
Performance Comparison

Model	Accuracy	Loss
Artificial neural network and Convolutional neural network	78 percent and 77 percent	1.5 and 1.3
Global model with LSTM and local models with LSTM and Autoencoder using Federated Learning	80 percent	0.7

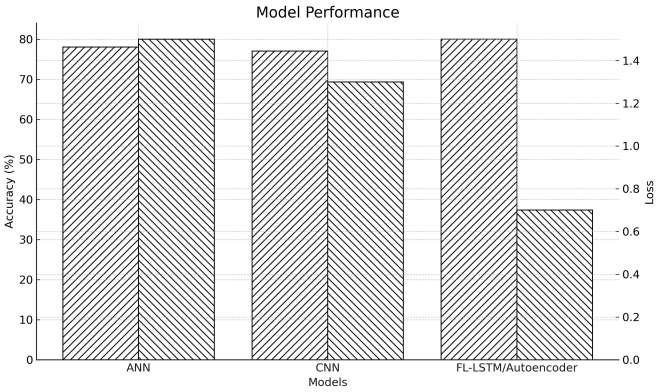

Fig. 3. Model Comparison

Through five communication rounds between clients and the global server, the resulting model resulted in 80 accuracy and a loss decrease of 0.7, improving upon earlier models while retaining data privacy with Federated Learning. The accuracy comparsion is shown in Figure 3. The Figure 3 compares the accuracy and loss of three different machine learning models: an Artificial Neural Network (ANN), a Convolutional Neural Network (CNN), and a Federated Learning Long Short-Term Memory/Autoencoder (FL-LSTM/Autoencoder). The accuracy is represented on the left y-axis as a percentage, while the loss is represented on the right y-axis.

From the graph, it can be observed that both the ANN and CNN models achieve relatively high accuracy, around 78-80

Specifically, the ANN and CNN models don’t have any loss shown. The FL-LSTM/Autoencoder shows the lowest loss, at around 0.7. This indicates that while all three models have similar accuracy, the FL-LSTM/Autoencoder is more confident in its predictions, making fewer mistakes and resulting in a lower loss value. This suggests the FL-LSTM/Autoencoder might be a better choice for the given task due to its higher confidence and overall better performance.

V. Conclusion

The model can be improved further by dividing the data in terms of attack classes instead of random distribution across clients. Here, the dataset is divided into 14 attack types and 4 big attack classes so that each client learns about

particular patterns of attacks. This approach improves local model evaluation individually so that each local model focuses on a specific set of attack behaviors. As a result, the federated model's overall performance is enhanced, with the combined expertise of specialized clients resulting in improved classification accuracy and a stronger Intrusion Detection System (IDS).

The further implementations of the model are splitting the data according to the attack class,here 14 classes of attacks and 4 main attacks are identified Rather than splitting data randomly among clients this kind of splitting may give more result for individual client evaluation,which leads to better performance

REFERENCES

[1] Sahbi, A., Jaidi, F., Bouhoula, A. (2023, June). Machine learning algorithms for enhancing intrusion detection within SDN/NFV. In 2023 International Wireless Communications and Mobile Computing (IWCMC) (pp. 602-607). IEEE.

[2] Prathibha, S., Bino, J., Ahammed, M. T., Das, C., Oion, S. R., Ghosh, S., Afroj, M. (2022, January). Detection methods for software defined networking intrusions (SDN). In 2022 International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI) (pp. 1-6). IEEE.

[3] Kumar, S., Kaswa, J., Meena, L., Porwal, M. (2022, June). Convolution neural network-based detection in software defined networks. In 2022 2nd International Conference on Intelligent Technologies (CONIT) (pp. 1-6). IEEE.

[4] Kumar, S., Kaswa, J., Meena, L., Porwal, M. (2022, June). Convolution neural network-based detection in software defined networks. In 2022 2nd International Conference on Intelligent Technologies (CONIT) (pp. 1-6). IEEE.

[5] Raza, M., Saeed, M. J., Riaz, M. B., Sattar, M. A. (2024). Federated Learning for Privacy Preserving Intrusion Detection in Software Defined Networks. IEEE Access.

[6] Lin, Z., Hongle, D. (2020, December). Research on SDN intrusion detection based on online ensemble learning algorithm. In 2020 International Conference on Networking and Network Applications (NaNA) (pp. 114-118). IEEE.

[7] Sarnovsky, M., Paralic, J. (2020). Hierarchical intrusion detection using machine learning and knowledge model. Symmetry, 12(2), 203.

[8] Zhao, Y., Chen, J., Wu, D., Teng, J., Yu, S. (2019, December). Multi-task network anomaly detection using federated learning. In Proceedings of the 10th international symposium on information and communication technology (pp. 273-279).

[9] Bhayo, J., Shah, S. A., Hameed, S., Ahmed, A., Nasir, J., Draheim, D. (2023). Towards a machine learning-based framework for DDOS attack detection in software-defined IoT (SD-IoT) networks. Engineering Applications of Artificial Intelligence, 123, 106432.

[10] Lee, T. H., Chang, L. H., Syu, C. W. (2020, June). Deep learning enabled intrusion detection and prevention system over SDN networks. In 2020 IEEE International Conference on Communications Workshops (ICC Workshops) (pp. 1-6). IEEE.

[11] Abd Al-Ameer, A. A., Bhaya, W. S. (2023). Enhanced Intrusion Detection in Software-Defined Networks Through Federated Learning and Deep Learning. Ingénierie des Systèmes d'Information, 28(5), 1213.

[12] Ma, X., Liao, L., Li, Z., Lai, R. X., Zhang, M. (2022). Applying federated learning in software-defined networks: A survey. Symmetry, 14(2), 195.

[13] de Oliveira, J. A., Gonçalves, V. P., Meneguette, R. I., de Sousa Jr, R. T., Guidoni, D. L., Oliveira, J. C., Rocha Filho, G. P. (2023). F-NIDS—A Network Intrusion Detection System based on federated learning. Computer Networks, 236, 110010.

[14] Cui, J., Sun, H., Zhong, H., Zhang, J., Wei, L., Bolodurina, I., He, D. (2023). Collaborative intrusion detection system for sdvn: A fairness federated deep learning approach. IEEE Transactions on Parallel and Distributed Systems.

[15] de Oliveira, J. A., Gonçalves, V. P., Meneguette, R. I., de Sousa Jr, R. T., Guidoni, D. L., Oliveira, J. C., Rocha Filho, G. P. (2023). F-NIDS—A Network Intrusion Detection System based on federated learning. Computer Networks, 236, 110010.

[16] Satheesh, N., Rathnamma, M. V., Rajeshkumar, G., Sagar, P. V., Dadheech, P., Dogiwal, S. R., ... Sengan, S. (2020). Flow-based anomaly intrusion detection using machine learning model with software defined networking for OpenFlow network. Microprocessors and Microsystems, 79, 103285.

[17] Ahmed, M. R., Shatabda, S., Islam, A. M., Robin, M. T. I. (2023). Intrusion Detection System in Software-Defined Networks Using Machine Learning and Deep Learning Techniques–A Comprehensive Survey. Authorea Preprints.

[18] Dawoud, A., Shahristani, S., Raun, C. (2018, May). A deep learning framework to enhance software defined networks security. In 2018 32nd International Conference on Advanced Information Networking and Applications Workshops (WAINA) (pp. 709-714). IEEE.

[19] Hande, Y., Muddana, A. (2019, November). Intrusion detection system using deep learning for software defined networks (SDN). In 2019 International Conference on Smart Systems and Inventive Technology (ICSSIT) (pp. 1014-1018). IEEE.

[20] Alzahrani, A. O., Alenazi, M. J. (2021). Designing a network intrusion detection system based on machine learning for software defined networks. Future Internet, 13(5), 111.